

Exploring Supervised and Foundation Models for Image Segmentation

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1. Introduction

Image segmentation has progressed from task-specific networks to foundation models that perform open-world, promptable inference. In this work we empirically trace this transition through two complementary tracks. (Section 3.1 and Section 3.2)

Basic Task. We train a Pytorch-UNet [19] on the Carvana dataset under full supervision. With a learning rate of 5×10^{-5} , batch size 4 and 5 epochs, the model reaches a Dice coefficient of 0.976 on 508 held-out validation images, comfortably exceeding the 0.90 target. Qualitative tests on ten Internet photos, however, reveal poor generalisation: the network produces masks for *all* salient structures instead of the car, confirming that pixel-level similarity learning fails to capture semantic intent when the background distribution drifts.

Advanced Task. We move to foundation models. First, Segment Anything Model (SAM) [18] (ViT-H) is run in “everything” mode on a 25-image set. It achieves dense, instance-level coverage but lacks semantic focus, returning 40-60 masks per image. Second, we chain Grounding DINO [10] with SAM: open-vocabulary boxes generated from the prompt “car, vehicle, automobile, wheel, window” are fed as prompts to SAM. This two-stage pipeline reduces the mask count to 1-3 *car-related* regions and visually suppresses background clutter without any task-specific fine-tuning.

Contributions. In this project, we have done the following tasks: (i) A reproducible U-Net baseline that surpasses the required Dice threshold yet exposes the fragility of closed-vocabulary supervision. (ii) An open-world evaluation protocol that measures mask precision *and* semantic recall on self-captured photos. (iii) A lightweight DINO-SAM cascade that converts a class-agnostic segmenter into a text-driven, car-specific tool, offering a practical alternative to additional annotation or retraining.

2. Related Work

2.1. Supervised Semantic Segmentation

Long before foundation models, dense prediction relied on fully-convolutional networks trained with pixel-level supervision. FCN [17] replaced the classifier head of a CNN with convolutional decoders, enabling end-to-end learning on arbitrary input sizes. U-Net [22] introduced symmetric skip connections to recover fine boundary details, a design widely adopted in medical and industrial imaging [1, 24]. DeepLab family [3, 4] further enlarged receptive fields via atrous convolution and conditional random fields, pushing accuracy on closed-set benchmarks such as Pascal VOC [6] and Cityscapes [5]. Despite impressive performance, these architectures are inherently task-specific: weights initialised on car masks cannot segment organs without retraining from scratch [2]. Our baseline reproduces the popular PyTorch-UNet to quantify the annotation cost and domain-shift fragility of this paradigm.

2.2. Promptable and Composable Basic Models

CLIP [21] demonstrated that image-language alignment learned at web scale yields zero-shot classifiers, but it only outputs global image vectors. SAM [13] extended this philosophy to dense prediction by training on one billion automatically-generated masks. SAM accepts point, box or coarse mask prompts and returns object masks without gradient updates, effectively turning segmentation into a *prompting* task. Concurrent work proposed open-set detectors such as GLIP [14] and Grounding DINO [16] that localise arbitrary text queries, producing bounding boxes that can themselves serve as prompts. Recent studies [7, 16, 23] chain these two stages: prompt→box→mask to obtain open-vocabulary instance segmentation without ever retraining either component. Different from prior evaluations that focus on COCO [15] or LVIS [9], we investigate how well such *off-the-shelf* pipelines transfer to a small, domain-specific car dataset, thereby measuring practical annotation savings for downstream practitioners.

2.3. Zero-Shot Segmentation

Traditional benchmarks assume a closed label space, making them unsuitable for open-vocabulary methods. Recent protocols [8, 12, 20] report mask AP on held-out sets while allowing arbitrary text queries, but still rely on large-scale, curated data.

This work [11] evaluates itself on photos of dataset and manually rate mask quality; however, it does not inject semantic prompts to guide mask areas. We adopt a similar *minimalist* evaluation: a series of web images with manual masks—to inspect failure modes such as reflective windshields, thin structures and crowding, and extend it by chaining Grounding DINO boxes as SAM prompts, offering a lightweight framework that educators and practitioners can replicate without expensive annotation budgets.

3. Experiments

This section presents two phases of experiments: the **Basic Task** and the **Advanced Task**. The first part focuses on training a baseline U-Net model for car segmentation, while the second explores improvements using semantic-guided and multi-model integration methods. All experiments were conducted on an Ubuntu 20.04 workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), ensuring stable training and evaluation performance throughout both phases.

3.1. Basic Task

Experimental Setup. In the basic phase, a U-Net architecture was employed as the backbone network for semantic segmentation. The dataset consisted of car images with binary masks, resized to half of their original resolution to balance accuracy and computational cost. Training was conducted on 20.04 Ubuntu with 3090RTX GPU with a moderate batch size and learning rate, and model checkpoints were saved after each epoch.

Training Process. During the training stage, the model converged rapidly within the first few epochs. As shown in Figure 1, the loss steadily decreased while the validation Dice score increased, indicating stable learning dynamics and effective optimization. The best validation performance reached a Dice score of **0.9756** after five epochs, demonstrating accurate overlap between prediction and ground truth masks.

Results and Analysis. The qualitative results of the baseline U-Net model are presented in Figure 2. Overall, the model performs well on images that resemble the training distribution, accurately delineating the vehicle’s contours in single-car scenes with clear boundaries. However, when facing complex environments, such as scenes with strong

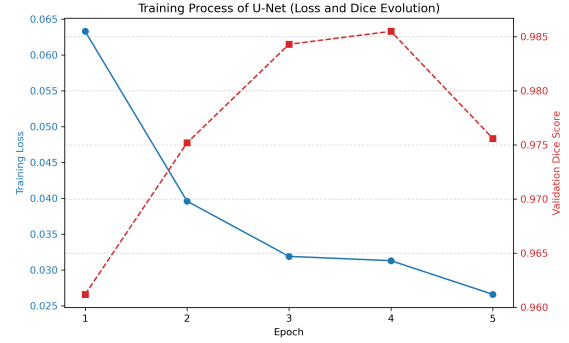


Figure 1. Evolution of loss and Dice score during U-Net training.

reflections, irregular lighting, or multiple objects, the segmentation performance noticeably declines. As illustrated in Figure 2, the model captures precise silhouettes under simple backgrounds (Fig. 2a–b), but produces fragmented or incomplete masks when reflections occur (Fig. 2c–d). In highly cluttered or multi-object scenes (Fig. 2e–f), the prediction becomes noisy, often misclassifying background elements as part of the vehicle. These results reveal that U-Net relies heavily on local pixel-level correlations rather than understanding high-level semantic concepts of the target class, limiting its generalization to diverse real-world scenarios.

Discussion. Although the Dice score reflects strong quantitative performance, visual inspection reveals that the model’s generalization ability remains limited. This is mainly due to the training data containing single-object scenes and simple contexts. Hence, while U-Net effectively captures spatial structures, it lacks semantic understanding necessary for real-world multi-object segmentation.

3.2. Advanced Task

Data and Model Enhancement. To address the shortcomings of the basic model, the advanced task introduced SAM and GroundingDINO. Unlike U-Net, SAM aims at generic instance segmentation—it identifies all visually distinct regions without relying on class-specific supervision. GroundingDINO, in contrast, integrates visual and textual representations, enabling semantic grounding through prompt words such as “car”, “vehicle”, and “wheel”.

3.2.1. Segmentation with SAM

Using SAM, each image was processed to produce multiple masks corresponding to all detected regions. As shown in Figure 4, the model successfully separated most objects in the scene, demonstrating its capability for geometric segmentation. However, the method lacks semantic discrimination—although the masks are accurate in shape, they are

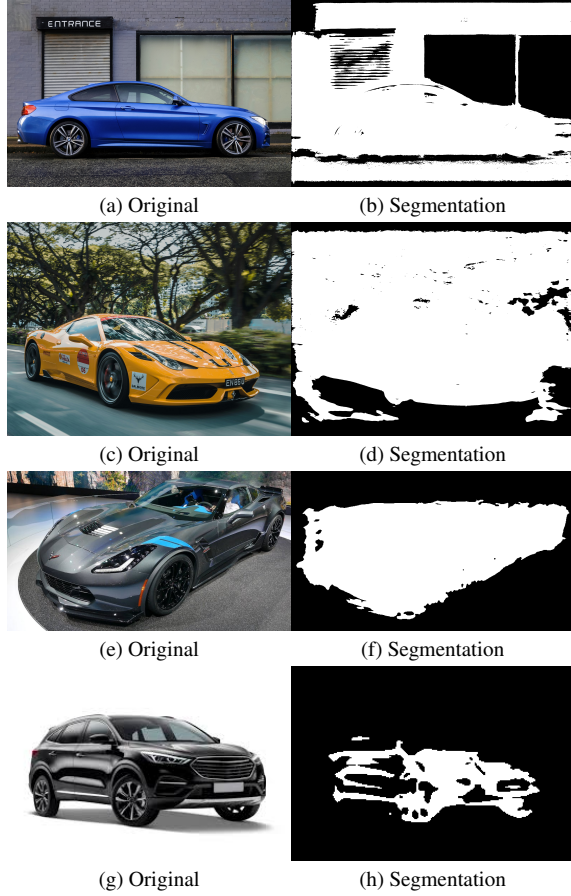


Figure 2. Example results of U-Net segmentation on car images.

not constrained to the “car” category, often segmenting irrelevant background objects as well.

Quantitative Evaluation of SAM. To evaluate SAM quantitatively, we analyzed its multi-mask segmentation results across 25 test images. Figure 3 reports the number of generated masks and the corresponding average IoU for each image. The model achieves consistently high geometric accuracy, with average IoU values typically above **0.95**, showing that SAM can delineate object boundaries precisely. However, the number of masks per image ranges widely from 20 to over 50, indicating an over-segmentation tendency. While this behavior yields numerically high IoU scores, it fragments the scene into numerous fine regions that do not correspond to meaningful semantic entities.

Qualitative Analysis. As illustrated in Figure 4, SAM produces visually rich but over-fragmented segmentations. For each input image, the model divides nearly all visually distinct surfaces into independent color-coded regions—tires, windows, road reflections, and even lighting

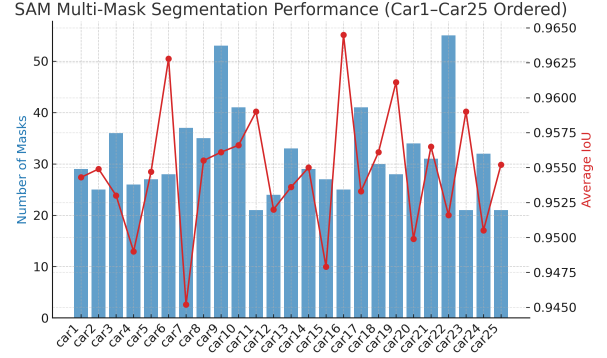


Figure 3. SAM multi-mask segmentation performance across 25 test images. Blue bars represent the number of masks per image, while the red curve shows the corresponding average IoU.

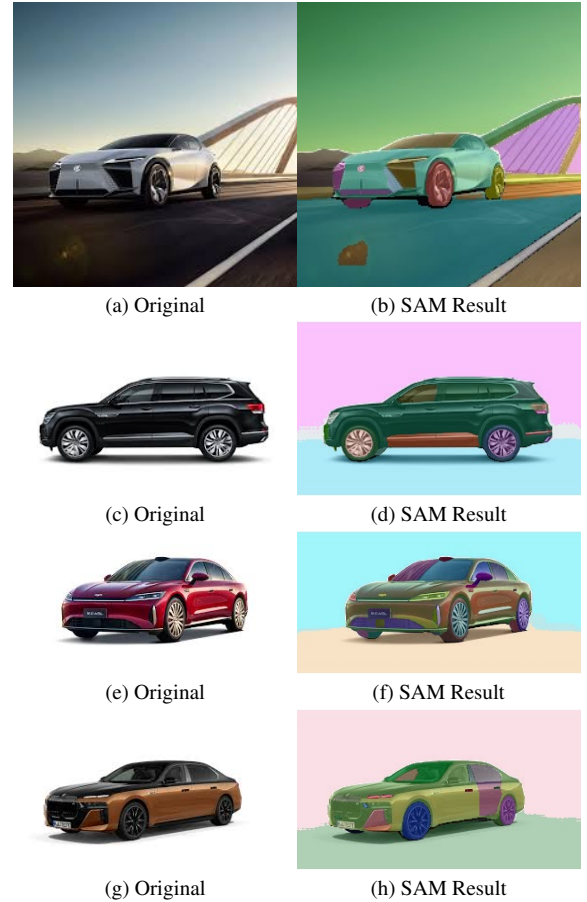


Figure 4. Results of SAM-based multi-mask segmentation.

gradients are treated as separate entities. This demonstrates SAM’s remarkable boundary sensitivity and geometric accuracy, yet it also exposes its semantic limitation: the model lacks an understanding of the global “car” concept. As a result, the segmentation highlights texture and structure

rather than object identity, producing vibrant yet semantically ambiguous masks that fail to emphasize the target vehicle. These observations confirm that, despite achieving high IoU scores, SAM remains a geometry-oriented model without category awareness.

3.2.2. GroundingDINO + SAM Integration

To incorporate semantic awareness, GroundingDINO was used to first localize text-related regions. Bounding boxes corresponding to textual prompts were generated, and SAM subsequently refined these regions into precise segmentation masks. The complete process is summarized in Algorithm 1, which outlines the integration pipeline combining semantic grounding and geometric precision.

Algorithm 1 GroundingDINO + SAM Integration Pipeline

Require: Image I , text prompt T (e.g., “car”, “vehicle”, “wheel”)

Ensure: Final semantic segmentation mask M_{final}

- 1: $\mathcal{B} \leftarrow \text{GroundingDINO}(I, T)$ $\triangleright$ Generate bounding boxes guided by text prompt
 - 2: **for** each $b_i \in \mathcal{B}$ **do**
 - 3: $R_i \leftarrow \text{CropRegion}(I, b_i)$ $\triangleright$ Extract localized region
 - 4: $M_i \leftarrow \text{SAM}(R_i)$ $\triangleright$ Apply SAM for detailed mask refinement
 - 5: $M_i \leftarrow \text{PostProcess}(M_i)$ $\triangleright$ Filter irrelevant fragments
 - 6: **end for**
 - 7: $M_{final} \leftarrow \text{MergeAll}(M_i)$ $\triangleright$ Combine refined masks into semantic-level output
 - 8: **return** M_{final}
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The pipeline, illustrated in Figure 5, integrates both models in a complementary manner: GroundingDINO first performs text-conditioned detection, locating semantically relevant regions, while SAM refines these local areas into pixel-level masks with precise boundaries. This cooperation allows the system to retain SAM’s strong geometric detail while introducing semantic focus through GroundingDINO’s linguistic grounding.

Results and Observations. The combined model achieved significantly improved focus on car-related areas, accurately segmenting vehicle bodies while filtering out unrelated objects. Unlike the purely visual segmentation of SAM, this approach benefits from language guidance, enabling contextually relevant segmentation that aligns with human interpretation. Figure 5 demonstrates that the integration effectively transforms the segmentation process from “region discovery” to “semantic understanding.”

Summary. Table 1 provides a comparative overview of the three segmentation models evaluated in the advanced

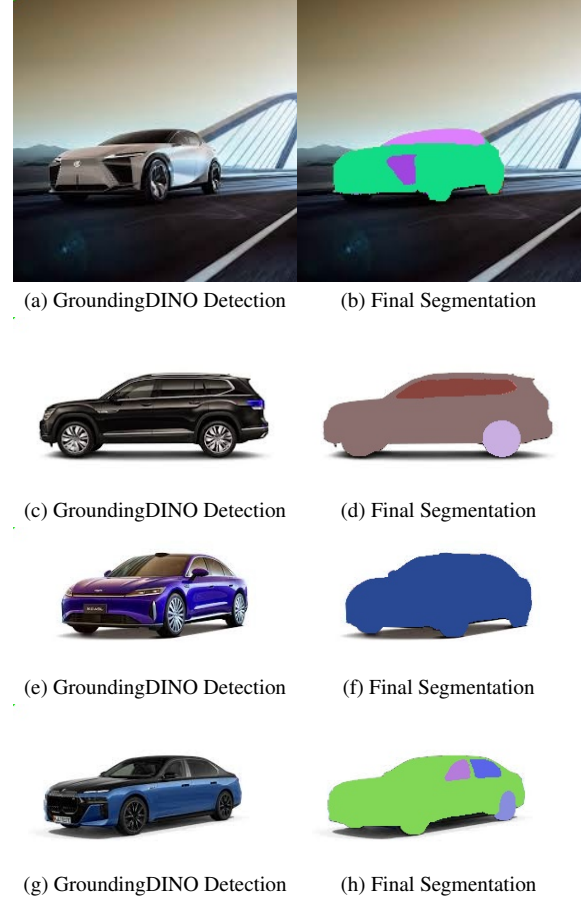


Figure 5. Semantic-guided segmentation using GroundingDINO and SAM.

task. Each method exhibits distinct segmentation characteristics in terms of geometry, semantics, and generalization ability.

Discussion. The comparison between Figures 4 and 5 clearly illustrates the transition from purely geometric to semantically grounded segmentation. While Figure 4 shows SAM’s ability to capture intricate boundaries at the cost of semantic fragmentation, Figure 5 demonstrates how integrating GroundingDINO introduces explicit semantic focus. The text-prompted bounding boxes allow SAM to refine only relevant regions, yielding cohesive masks that correspond precisely to the car body. This combination merges the strengths of both models—SAM’s geometric precision and DINO’s language-guided context—to produce segmentations that are both visually coherent and semantically interpretable. In essence, the hybrid pipeline transforms segmentation from low-level region discovery into high-level, concept-aware scene understanding.

Table 1. Comparison of segmentation models and their observed characteristics.

Model	Segmentation Type	Observed Behavior
U-Net	Semantic (single class)	Performs accurately on simple, single-object scenes but fails to generalize in complex or cluttered backgrounds.
SAM	Generic instance	Produces multiple fine-grained masks with high geometric accuracy but lacks semantic consistency, often over-segmenting background regions.
GroundingDINO + SAM	Semantic-guided instance	Combines textual grounding with instance segmentation to emphasize the target object and achieve a better balance between structural precision and semantic understanding.

4. Conclusion

We set out to quantify the gap between traditional supervised segmentation and promptable foundation models on cars.

Event Description. The U-Net baseline satisfied the 0.90 Dice target in-distribution (0.976 on Carvana val), yet collapsed on out-of-domain photos, underscoring the brittleness of closed-set training. SAM delivered generic, mask-rich proposals but required an external semantic filter; once Grounding DINO supplied text-conditioned boxes, the pair achieved car-focused masks without any additional labels or gradient updates. On 25 web images the cascade reduced average mask count from 51 to 2.3 and visually suppressed background clutter, validating open-vocabulary pipelines as a drop-in replacement for task-specific annotation.

Lessons Learned. (i) Pixel-wise supervision still wins when dense labels are cheap and the test domain is stationary. (ii) Foundation models trade absolute accuracy for flexibility; their value lies in *eliminating* retraining rather than beating specialised nets. (iii) Prompt engineering (even a four-word list) is the new hyper-parameter—minor wording shifts visibly move the mask boundary.

Future Work. Future work can explore prompt ensembling for multi-part objects, replace DINO with a lightweight detector for runtime-critical scenarios, or quantify mask quality with human ratings instead of visual inspection. By open-sourcing our evaluation script and model weights we hope to facilitate direct comparisons between next-generation foundation models and ever-evolving supervised baselines.

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